



A 21-Color Panel Designed for the ZE5 Cell Analyzer to Enumerate Rare Stem Cell Populations in Human Cord Blood

Michelle Scott, Christine Thornton, and Matt Alexander
Bio-Rad Laboratories, Inc., Hercules, CA 94547 USA



Introduction

The Bio-Rad ZE5 Cell Analyzer is a tabletop flow cytometer configurable with up to five lasers and 28 photomultiplier tube (PMT) fluorescence sensors. The integrated Everest Software suite incorporates a large fluorophore library mapped to the instrument’s optimal laser excitation and emission filter sensor to make complex multicolor panels easier to design and input when creating experiments. Along with a semiautomated compensation routine, Everest Software will help guide the researcher in setting up proper single-color controls for acquisition and automatic compensation matrix generation. In this poster we will demonstrate the necessary steps in setting up a 21-color panel on the ZE5 using single-color control beads for autocompensation generation of a 21-color compensation matrix. We will then export the compensated FCS files for complex subset analysis using FlowJo Analysis Software to enumerate most major populations present in normal cord blood and demonstrate the reliability of enumerating rare stem cell populations present in the same sample (Keeney and Sutherland 2000). Complex multicolor cytometry panels are becoming more common and desired when trying to understand the relationship of rare cell populations present in relatively precious samples (Brocklebank and Sparrow 2001). The ZE5 Cell Analyzer makes this easier to accomplish.

Materials and Methods

Antibodies: CD45-BUV395, CD19-BUV496, CD4-BUV661, CD25-BUV737, CD3–Brilliant Violet 421, CD38-PacBlu, CD135-BV510, CD29-BV605, CD14-BV650, CD45RA-BV711, CD27-BV785, CD44-BB515, CD11b-PerCP, CD73-BB700, CD34-PE, CD105-PE-CF594, CD127-PE-Cy5, CD10-PE-Cy7, CD90-APC, CD71-AF700, DRAQ7

Blocking buffer: PBS + 50 µg/ml mouse, rat, rabbit IgG + 1% BSA + 1% glucose + 20 mM EDTA + 0.1% azide

Washing buffer: PBS + 1% BSA + 1% glucose + 0.1% azide

Compensation beads: Quantum Simply Cellular Anti-Mouse (catalog #815) and Anti-Rat (#817) Capture Beads (Bangs Laboratories, Inc.)

Media: RPMI 1640 + 10% FBS

Procedure

Cell preparation: Human cord blood mononuclear cells were thawed and stained using standard procedures. Thawed samples were placed in 5 ml of prewarmed cell media in 50 ml polypropylene centrifuge tubes. Cells were counted and assessed for viability using Bio-Rad’s TC20 Automated Cell Counter to obtain an accurate cell count and viability for each sample. Cord blood mononuclear cells were allowed to recover in culture media in a 37°C, 5% CO₂ incubator for at least an hour.

Cell staining: The cord blood mononuclear cells were treated with blocking buffer for 10 min, then stained by adding a prepared cocktail containing all 20 antibodies. Samples were incubated on ice, protected from light, for 45 min to an hour. Samples were washed twice with ice-cold wash buffer and resuspended in 1 ml of ice-cold wash buffer containing DRAQ7. Samples were stored on ice and protected from light until acquisition on the ZE5 Cell Analyzer. Stained compensation beads were used for single-stained controls.

Antibodies: All antibodies were run individually against cord blood, using eight-fold dilutions, in order to find the nominal titer for antibody staining of cord blood.

Compensation: Rather than using single-level antibody capture beads for compensation determination, Quantum Simply Cellular Anti-Mouse Beads, which capture four increasing levels of antibody, plus an unlabeled bead were used. This results in five distinct levels of signal that are used to generate a matrix. By doing so, subtle channels of overcompensation and undercompensation can be easily visualized and corrected manually in the resulting compensation matrix.

Results

1.4 x 10⁶ cells were acquired for analysis. After autogeneration of compensation matrix using Quantum Simply Cellular Anti-Mouse Capture Beads, the FCS 3.1 files along with the compensation matrix were downloaded to FlowJo Analysis Software v10 (FlowJo, LLC). Using the FlowJo compensation matrix editor, areas of over and undercompensation were corrected and subsequently used for all compensated data.

An unstained population of cord blood was used to set PMT voltages for all parameters such that all populations were well above signal background levels. Single-color stained cells and fluorescence minus one (FMO) controls were used to set gates for subsequent analysis. The initial gating scheme consisted of eliminating debris, nonviable cells, and cells positive for CD45. From the initial gates, a forward vs. side scatter bivariate plot was generated to display small lymphoid cells from larger myeloid cells. A gate was drawn around the lymphoid population to capture CD34+ cells and used to break down subpopulations and enumerate hematopoietic stem cells (HSC): 0.01% of all live cells, multipotent progenitor (MPP) cells: 0.16% of all live cells, and common lymphoid progenitor (CLP) cells: 0.05% of all live cells. Figure 1 illustrates the gating scheme by which the stem and progenitor cells were identified (HSC = CD34+CD38-CD90+CD45RA-CD45^{low}, MPP = CD34+CD38-CD90-CD45RA-CD45^{low}, CLP = CD34+CD10+CD45+).

Other plots were generated to demonstrate the representative populations of CD3+ T cells, CD19+ B cells, etc. present within the same stained sample. Figure 2 demonstrates some of the other expected populations (CD19+ B cells, CD3+ T cells, CD3+CD4+ T-helper cells, etc.) (Barnett et al. 2000, Chang and Ma 1996, Gratama 1998).

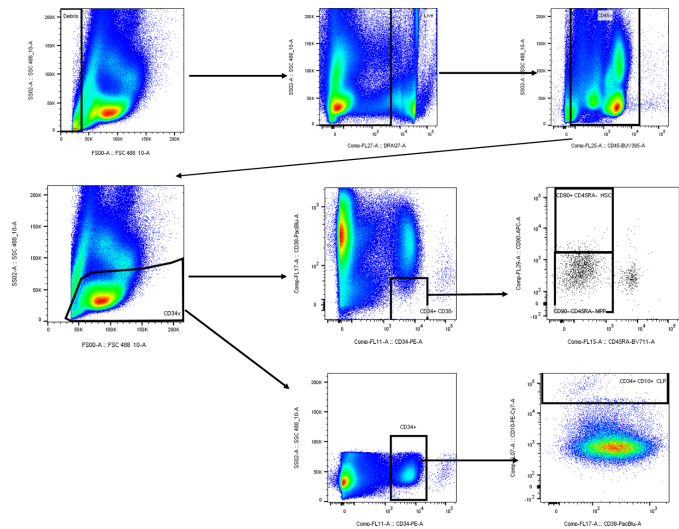


Fig. 1. Gating scheme to enumerate stem cell populations. HSC = 0.01%, MPP = 0.16%, and CLP = 0.05%.

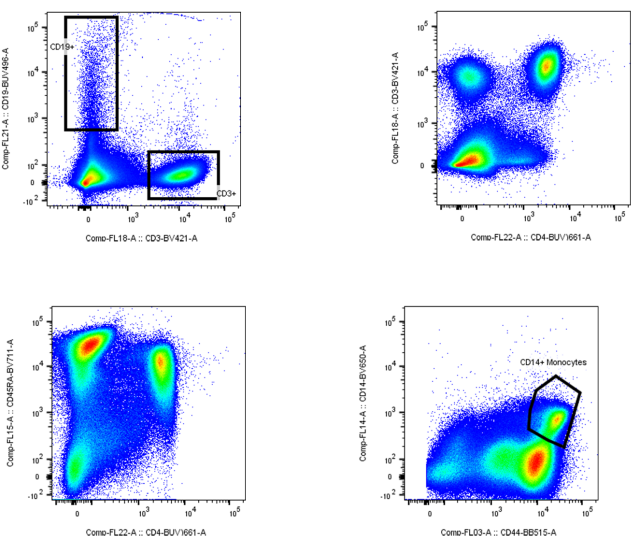


Fig. 2. Plots demonstrating presence of other expected populations, including B cells, T cells, and monocytes.

Markers

Marker	Other Nomenclature	Function
CD45	Leukocyte common antigen (LCA)	Regulator of T- and B-cell antigen receptor signaling; regulator of cell growth and differentiation
CD19	B4	Assembles with the antigen receptor of B lymphocytes to decrease the threshold for antigen receptor–dependent stimulation
CD4	L3T4, W3/25, T4	Initiates or augments the early phase of T-cell activation
CD25	TAC antigen, IL-2Rα, p55, TCGFR	Receptor for interleukin-2
CD3	T3	A complex of subunits that mediates T-cell signal transduction
CD38	ADP-ribosyl cyclase, T10, cyclic ADP–ribose hydrolase 1	Cell adhesion and signal transduction
CD135	FLT3, FLK2, STK1	Signaling through CD135 plays a role in cell survival, proliferation, and differentiation. CD135 is important for lymphocyte (B cell and T cell) development, but not for the development of other blood cells (myeloid development)
CD29	Platelet GPIIa, integrin β1, GP	Cell adhesion
CD14	LPS receptor	Mediates the innate immune response to bacterial lipopolysaccharide (LPS)
CD45RA	PTPRC	CD45 isoform
CD27	T14, S152, TP55, TNFRSF7	Generation and long-term maintenance of T-cell immunity
CD44	ECMR1I, H-CAM, Pgp-1, phagocytic glycoprotein I, extracellular matrix receptor III, GP90 lymphocyte homing/ adhesion receptor, hyaluronate receptor	Cell adhesion and migration
CD11b	Integrin αM, CR3, Mo1, C3niR, Mac-1	Implicated in the various adhesive interactions of monocytes, macrophages, and granulocytes
CD73	Ecto-5'-nucleotidase, NT5E, E5NT, NT5, NTE, eN, eNT	An ecto-5'-nucleotidase that hydrolyzes extracellular nucleotides into membrane permeable nucleosides
CD34	gp105-120, hematopoietic progenitor cell antigen 1 (HPCA1)	Cell adhesion
CD105	Endoglin, HHT1, ORW, SH-2	May play a role in hematopoiesis and angiogenesis
CD127	p90, IL-7R, IL-7Rα	Receptor for IL-7 and thymic stromal lymphopoietin (TSLP). This receptor is important for V(D)J recombination during development
CD10	CALLA, NEP, gp100, EC 3.4.24.11, MME	Neutral endopeptidase that cleaves peptides and inactivates several peptide hormones
CD90	Thy-1	Cell adhesion
CD71	TFRC, T9, transferrin receptor, TFR, TRFR	Mediates the uptake of transferrin-iron complexes
DRAQ7		Used for differentiating live and dead cells

References

Barnett D et al. (2000). Reduction of intra- and interlaboratory variation in CD34+ stem cell enumeration using a stable test material, standard protocols and targeted training. Br J Haematol 108, 784–792.

Brocklebank AM and Sparrow RL (2001). Enumeration of CD34+ cells in cord blood: A variation on a single-platform flow cytometric method based on the ISHAGE gating strategy. Cytometry 46, 254–261.

Chang A and Ma DD (1996). The influence of flow cytometric gating strategy on the standardization of CD34+ cell quantitation: An Australian multicenter study. J Hematother 5, 605–616.

Gratama JW et al. (1998). Flow cytometric enumeration of CD34+ hematopoietic stem and progenitor cells. Cytometry 34, 128–142.

Keeney M and Sutherland DR (2000). Stem cell enumeration by flow cytometry: Current concepts and recent developments in CD34+ cell enumeration. Cytotherapy 2, 395–402.

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